Machine Learning Methods for n-Colorant Printer Characterization: A Corrected and Extended Evaluation

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Abstract

[TODO: Abstract stub – fill in once Results/Discussion are drafted (Task 2). MDPI Technologies caps the abstract at 200 words; single paragraph, no citations, no undefined abbreviations.] Printer color characterization maps a device's colorant inputs (CMY, CMYK, or higher-channel combinations) to device-independent color coordinates (CIE XYZ/Lab). A prior conference study (AIC 2025) compared 14 machine learning methods against third-order polynomial regression for $n = 3$ (CMY) inputs across three printer datasets, finding no consistent advantage for the learned models. That analysis is superseded here: its CIEDE2000 errors were computed on normalized rather than physical XYZ values, understating errors by roughly a factor of five. This paper reports corrected $n = 3$ results, extends the comparison to $n = 4$ (CMYK) inputs, **[RESULTS-PENDING: n>4 (CMYKOGV) results, if the dataset lands in time]**, and **[RESULTS-PENDING: cross-printer / newsprint generalization and direct-ΔE_{00}-loss results, if those experiments land in time]**. **[TODO: One-sentence headline finding once available.]**

Keywords: printer characterization; color management; CIEDE2000; machine learning; Gaussian process regression; polynomial regression; CMYK; n-colorant printing; ICC profiling

1 Introduction

[TODO: Extend the AIC 2025 introduction (Task 2, per docs/plans/07-paper.md). Planned additions: (i) the lookup-table-inefficiency motivation for higher-channel printing (Phil, Apr 2026 meeting) – LUTs scale combinatorially with the number of colorants, so device characterization methods that avoid dense LUT construction become increasingly attractive as n grows beyond 4; (ii) restate Phil Green's two core questions verbatim: (a) can ML/AI match or beat existing methods for $n \leq 4$ colorants, avoiding the need for gray-component-replacement-style algorithms; (b) can AI handle $n > 4$ colorant systems (e.g. CMYKOGV) where classical polynomial methods are not designed to scale; (iii) state explicitly that this paper corrects and supersedes the AIC 2025 conference numbers (forward-reference to the Methods correction paragraph).]

2 Related Work

[TODO: Task 2. Prior device-characterization literature already in refs.bib (Bala 2003; Cheung et al. 2004; Kang & Anderson 1992; Tominaga 1993) plus the per-method ML references (Breiman 1984/2001; Cortes & Vapnik 1995; Cover & Hart 1967; Friedman 2001; Geladi & Kowalski 1986; Hoerl & Kennard 1970; Rasmussen & Williams 2006; Tibshirani 1996; Zou & Hastie 2005). Add: Kiran's Optics Express

n-colour paper and the PhD-work benchmark Phil pointed to (both currently commented placeholders in refs.bib – exact citations pending); ISO/TC 130 standardization context (Pang 2024); the FOGRA51 characterization dataset (Fogra 2020); the colour-science library used throughout this pipeline (placeholder in refs.bib, pending exact version/DOI).]

3 Methods

3.1 Datasets

Three printer characterization datasets are used, each measured at 1,617 CMYK combinations under a shared measurement protocol (spectrophotometric CIE Lab and CIE XYZ, D50 illuminant, 2° standard observer). Each dataset is evaluated twice: once restricted to the $K = 0$ subset with only C, M, Y as inputs (818 rows – the $n = 3$ / CMY condition), and once using the full sample set with C, M, Y, K as four inputs (1,617 rows – the $n = 4$ / CMYK condition). Unlike the AIC 2025 conference version, the K channel is never dropped while its rows are kept: doing so maps identical CMY inputs to different XYZ outputs whenever K differs, which is an internal label inconsistency, not a modeling choice.

Table 1: Datasets used in the $n = 3$ and $n = 4$ conditions. Row counts are pooled across all 5 cross-validation folds (Section 3.3), so every sample is predicted exactly once.

Dataset	Substrate	Inputs (n)	Rows	Source
PC10	Cardboard	CMY	818	APTEC PC10 (2023)
PC10	Cardboard	CMYK	1,617	APTEC PC10 (2023)
PC11	Coated paper (CCNB)	CMY	818	APTEC PC11 (2023)
PC11	Coated paper (CCNB)	CMYK	1,617	APTEC PC11 (2023)
FOGRA51	Reference (ISO 12647-2)	CMY	818	Fogra Research Institute [1]
FOGRA51	Reference (ISO 12647-2)	CMYK	1,617	Fogra Research Institute [1]

[TODO: Task 2/06: append IFRA newsprint (white-backing / black-backing, 43 press runs, 1,485 samples each, 36-band spectral → XYZ via colour-science) once plan 03 lands, keeping wb and bb as separate rows per Phil's instruction. Append CMYKOGV ($n = 7$) row if that dataset arrives from Phil/Will in time (plan 06).]

3.2 Models

Fourteen regression methods are compared, matching the AIC 2025 study's model set with two configuration corrections noted below. All models are implemented input-dimension-agnostically, so the same code path serves the $n = 3$, $n = 4$, and (if it lands) $n = 7$ conditions.

[TODO: Task 2: add explicit equations for each model, in particular the corrected 3rd-order polynomial expansion (Phil flagged an "odd expression" in the AIC version's equation to fix). Every stochastic model above is seeded (fixed seed = 42) for reproducibility, per CLAUDE.md's pinned-dependency rule.]

3.3 Evaluation Protocol

Each (dataset, model) pair is evaluated with 5-fold cross-validation (fixed seed = 42). Folds are grouped by distinct input recipe (rounded to 6 decimal places) so that duplicate rows sharing the same colorant input always fall in the same fold, preventing train/test leakage through repeated measurements. Within each fold:

Table 2: Model registry (14 methods). Configurations match the AIC 2025 paper's best configs except where marked (*), which were degenerate there (Lasso/Elastic Net at $\alpha = 1.0$ collapsed to a constant predictor; SVR at $\varepsilon = 0.1$ produced an insensitivity tube covering 10% of the target range) and are replaced with standard, non-degenerate values.

Model	Configuration
Polynomial regression (degree 3)	Full 3rd-order feature expansion + ordinary least squares
Ridge regression	$\alpha = 0.5$
Lasso regression*	$\alpha = 10^{-3}$
Elastic Net*	$\alpha = 10^{-3}$, ℓ_1 ratio $= 0.5$
Principal component regression	All components retained + Ridge ($\alpha = 0.5$)
Partial least squares regression	3 components
k-nearest neighbors	$k = 5$, uniform weights
Support vector regression*	RBF kernel, $C = 10$, $\gamma =$ scale, $\varepsilon = 0.01$
Decision tree	Unconstrained depth
Random forest	200 trees, max depth 15
Gradient boosting	200 estimators, learning rate 0.05, max depth 5
Gaussian process regression	Const $\times$ RBF + WhiteKernel(10^{-5}), y-normalized
MLP (shallow)	1 hidden layer, 64 units, L-BFGS
MLP (deep)	3 hidden layers, 64 units each, L-BFGS

1. A `MinMax` scaler is fit on the training fold's inputs and, separately, on the training fold's XYZ targets – scaling parameters are never estimated from the held-out fold.

2. The model is fit in the normalized (0–1) input/output space.

3. Predictions are generated in normalized space, then inverse-transformed back to physical XYZ using the training fold's target scaler.

4. Predicted XYZ values are clipped at zero (tristimulus values cannot be negative).

5. Both predicted and ground-truth XYZ are converted to CIE Lab under the D50 illuminant and 2° standard observer, and the per-sample color difference is computed with the CIE 2000 formula [2] via the `colour-science` library.

Every sample in a dataset is held out exactly once across the 5 folds, so the reported statistics pool over the entire dataset rather than a single train/test split. Per repository convention (see CLAUDE.md), summary statistics are the **median**, **95th percentile**, and **maximum** of the per-sample ΔE_{00} distribution – not the mean or standard deviation, since color-difference errors are not normally distributed and a small number of large outliers otherwise dominate a mean-based summary.

As a data-integrity check, every dataset's own measured XYZ is round-tripped through the same D50/2°/CIE-2000 conversion and compared against its own measured Lab before any model is fit; a median absolute Lab discrepancy above 0.6 aborts the run. This tripwire is a direct response to the scale error described below, and guards against future recurrences (e.g. accidentally feeding normalized XYZ into the Lab conversion).

Correction to the AIC 2025 conference results. The AIC 2025 conference version of this analysis computed ΔE_{00} on normalized (0–1 scaled) XYZ values rather than on the physical 0–100 XYZ scale, which understated the reported color differences by approximately a factor of five. All results reported in this paper are corrected and supersede the conference figures.

4 Results

4.1 $n = 3$ (CMY) – Corrected

Table 3 reports median, 95th-percentile, and maximum ΔE_{00} for all fourteen models on the three $n = 3$ (CMY) datasets, pooled across the 5 cross-validation folds so that every one of the 818 $K = 0$ samples per dataset is predicted exactly once. Figure 1 shows the same medians as a ranked bar chart across the three datasets on a log scale, with a dashed reference line at $\Delta E_{00} = 1$ marking the conventional just-noticeable-difference threshold. Gaussian process regression is the best-median model on all three datasets by a wide margin, followed by third-order polynomial regression as a clear second; the ridge, PCR, PLSR, lasso, and elastic-net family cluster together with median ΔE_{00} around 6.3–6.7, an order of magnitude worse than GP and roughly 20× the just-noticeable-difference threshold.

Table 3: Median, 95th-percentile, and maximum ΔE_{00} per model for the $n = 3$ (CMY) condition on the three printer datasets, 818 $K = 0$ samples each. All values from 5-fold cross-validation with every sample predicted exactly once; ΔE_{00} computed on denormalized XYZ (D50 illuminant, 2° standard observer), per Section 3.3. Models are sorted by PC10 median. Source: `journal/results/{PC10,PC11,FOGRA51}-CMY/summary.csv`.

	PC10			PC11			FOGRA51		
Model	Median	P95	Max	Median	P95	Max	Median	P95	Max
Gaussian Process	0.054	0.189	2.061	0.054	0.160	1.797	0.065	0.191	0.807
Polynomial (3rd order)	0.279	0.966	7.695	0.244	0.890	7.107	0.368	1.163	8.008
SVM	0.754	1.947	5.442	0.717	1.836	4.889	0.766	1.838	6.168
Gradient Boosting	0.881	3.133	22.659	0.791	2.821	13.192	0.819	2.926	17.404
MLP (Deep)	1.102	4.055	15.958	1.054	3.736	25.396	1.176	4.051	23.968
MLP (Shallow)	1.195	4.483	21.150	1.176	4.001	13.442	1.408	6.086	18.983
Random Forest	1.716	4.187	9.264	1.694	4.108	8.467	1.804	4.486	8.131
k-NN	2.012	4.270	8.918	1.938	4.085	8.852	1.867	4.345	9.512
Decision Tree	4.369	9.828	25.211	4.136	8.891	24.132	4.320	9.001	20.566
Lasso	6.601	24.791	43.961	6.292	23.462	45.338	6.658	25.221	42.062
Elastic Net	6.611	24.781	43.890	6.284	23.952	45.261	6.692	25.279	41.981
Ridge	6.624	24.802	43.820	6.309	23.943	45.184	6.719	25.397	41.897
PCR	6.624	24.802	43.820	6.309	23.943	45.184	6.719	25.397	41.897
PLSR	6.643	24.901	43.823	6.307	23.989	45.184	6.715	25.528	41.892

4.2 $n = 4$ (CMYK) and the n=3 → n=4 Comparison

[RESULTS-PENDING: Table: median/P95/max ΔE_{00} per model per dataset (*-CMYK). Source: `journal/results/{PC10,PC11,FOGRA51}-CMYK/summary.csv`.]

Figure 2 compares median ΔE_{00} at $n = 3$ (CMY) versus $n = 4$ (CMYK) for every model on PC10, the full 1,617-sample set now including the $K > 0$ rows. Adding the fourth input channel is not uniformly costly: third-order polynomial regression, which has no mechanism to represent the extra channel's interaction terms beyond its fixed cubic expansion, degrades from a median of 0.279 to 0.944 (×3.4), whereas Gaussian process regression is comparatively robust, degrading only from 0.054 to 0.063 (×1.2). This asymmetry is consistent with GP regression's non-parametric kernel adapting to the additional input dimension, while the fixed-order polynomial basis cannot.

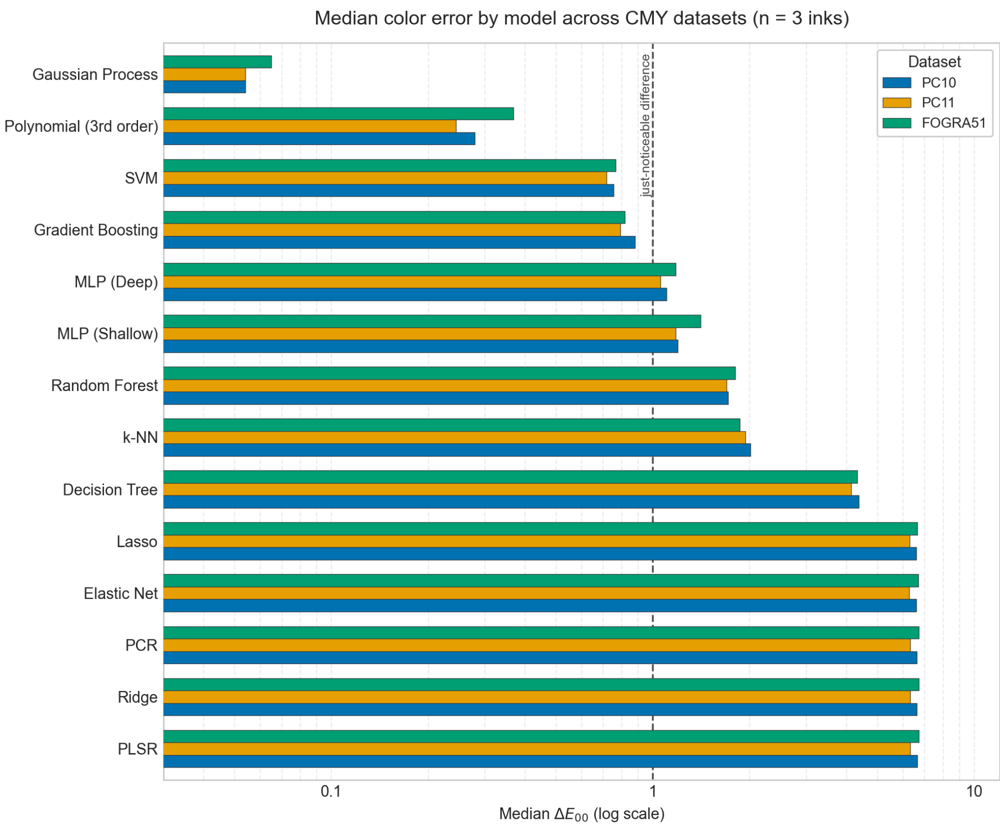

Figure 1: Median ΔE_{00} per model across the three CMY datasets (818 $K = 0$ samples each, 5-fold cross-validation), ranked by mean median across datasets. Log scale; the dashed line marks the just-noticeable difference ($\Delta E_{00} \approx 1$).

4.3 Gaussian Process vs. Noise Floor

Gaussian process regression is the best-median model in every dataset/condition combination reported above. Because GP is also the model most exposed to a subtle cross-validation leak – duplicate colorant-input rows with distinct measured XYZ landing in both the train and test folds – its result was re-verified with a duplicate-grouped cross-validation split (folds assigned by distinct input recipe rather than by row, so all duplicates of a given recipe fall in the same fold; see Section 3.3). Across all six dataset/condition combinations, GP's median ΔE_{00} ranges from 0.052 to 0.073, and the grouped-CV median never differs from the plain k-fold median by more than a factor of 1.02, confirming the result is not an artifact of duplicate leakage. Third-order polynomial regression, re-checked the same way, is similarly stable. Source: `journal/results/gp_verification.csv`.

4.4 IFRA Newsprint: Within-Run, Cross-Run, and Leave-One-Out

[RESULTS-PENDING: Plan 03 results, conditional on ingestion completing. Source: `journal/results/ifra/*.csv`. Figure F4.]

4.5 Direct ΔE_{00} Loss

[RESULTS-PENDING: Plan 05 results, conditional on landing in time.]

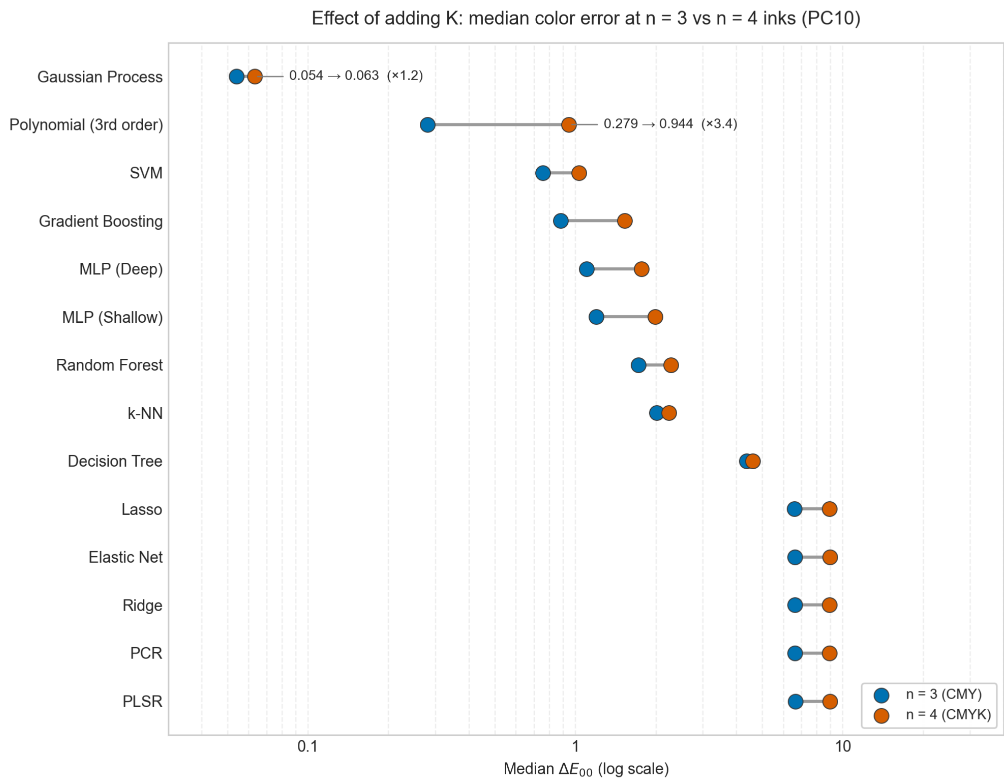

Figure 2: Median ΔE_{00} per model on PC10 at $n = 3$ (CMY, 818 samples) versus $n = 4$ (CMYK, 1,617 samples). Polynomial regression degrades $\times 3.4$ with the added K channel, while Gaussian process regression degrades only $\times 1.2$, the two highlighted comparisons.

4.6 LLM-as-Color-Predictor

[RESULTS-PENDING: Plan 04 results, conditional on landing in time. Source: `journal/results/llm/comparison.csv`. Figure F5.]

4.7 $n = 7$ (CMYKOGV)

[RESULTS-PENDING: Plan 06 results, conditional on the dataset arriving from Phil in time. Figure F6 (median ΔE_{00} vs. $n \in \{3, 4, 7\}$).]

5 Discussion

[TODO: Task 2. Answer Phil's two core questions explicitly: (a) does any ML method beat classical polynomial regression at $n \leq 4$, and if so by how much, avoiding gray-component-replacement-style handling of K; (b) does any method handle $n > 4$ successfully where polynomial regression is not designed to scale (contingent on the $n = 7$ dataset). Limitations to cover: single-substrate training sets (no cross-printer generalization claim without the IFRA results); GP's $O(N^3)$ training-set-size scaling as a practical deployment concern; the K-ambiguity lesson from the AIC 2025 conference version as a cautionary methodological note (kept minimal, per author decision – see Methods correction paragraph – no separate pitfalls section unless Phil asks for one on review).]

6 Conclusion

[TODO: Task 2. One paragraph, restating the corrected headline findings and, if landed, the $n > 4$ finding.]

Data Availability Statement

Code and data supporting this study are available at https://github.com/hamzafer/Evaluating-Machine-Lea **[TODO: Confirm this is still the correct public repository URL before submission (ported from the AIC 2025 conference paper's Code and Data Availability section) and that it reflects the corrected pipeline described here, not only the legacy AIC v1 code.]**

References

[1] Fogra Research Institute. FOGRA51 and FOGRA52 characterisation data (version 1.0) [data set]. https://fogra.org/en/downloads/work-tools/characterisation-data, 2020.

[2] Gaurav Sharma, Wencheng Wu, and Edul N. Dalal. The CIEDE2000 color-difference formula: Implementation notes, supplementary test data, and mathematical observations. *Color Research & Application*, 30(1):21–30, 2005.